

Systems Analysis & Control: Project I

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Introduction:

This project focuses on the modelling and analysis of a vertically landing rocket system. A non-linear model is derived and describes the translational and rotational motion of the rocket, accounting for aerodynamic forces such as lift, thrust, drag, and gravity. The system is non-linear due to several terms dependant on non-constant velocities, and trigonometric relations.

The purpose of the project is to derive the non-linear modelling equations of a system describing the final part of the flight path for a self-landing rocket. During this landing procedure, we find the system's equilibriums and derive new linearized approximations about this point. This enables us to better analyse the system to work with a simpler model that still effectively approximates the real-life behaviour near the equilibrium point.

The system behaviour is analysed in terms of its stability and response, which were characterised by the angles of rotation, as well as the forces acting on the rocket and its mass:

Signal	Type	Symbol	Units
Horizontal position	State	$x(t)$	m
Horizontal velocity	State	$\dot{x}(t)$	m/s
Vertical position	State	$y(t)$	m
Vertical velocity	State	$\dot{y}(t)$	m/s
Rocket angle	State	$\theta(t)$	deg
Rocket angular velocity	State	$\dot{\theta}(t)$	deg/s
Engine gimbal angle	Input	$u(t)$	deg

Table 1: Relevant signals for the Falcon 9 rocket.

Parameter	Symbol	Value	Units
Dry mass (excluding fuel)	m_0	25,600	kg
Initial mass of fuel	m_f	5,000	kg
Fuel burn rate at full throttle	$\dot{m}$	-300	kg/s
Height of center of mass (c.m.)	L_{cm}	12	m
Total height	L_0	47.7	m
Diameter of the rocket	D_0	3.66	m
Moment of inertia about the c.m.	J	4,400,000	kg m ²
Main engine thrust force (full throttle)	T_{max}	845,000	N

Table 2: Numerical parameter values for first stage of Falcon 9.

Name	Symbol	value
Axial area	A_a	$\frac{1}{4} \pi D_0^2$
Normal area	A_n	$L_0 D_0$
Axial drag coefficient	C_a	0.82
Normal drag coefficient	C_n	1.2
Density of air	ρ	1.225 kg/m ³ (at sea level)

Table 3: Aerodynamic coefficients.

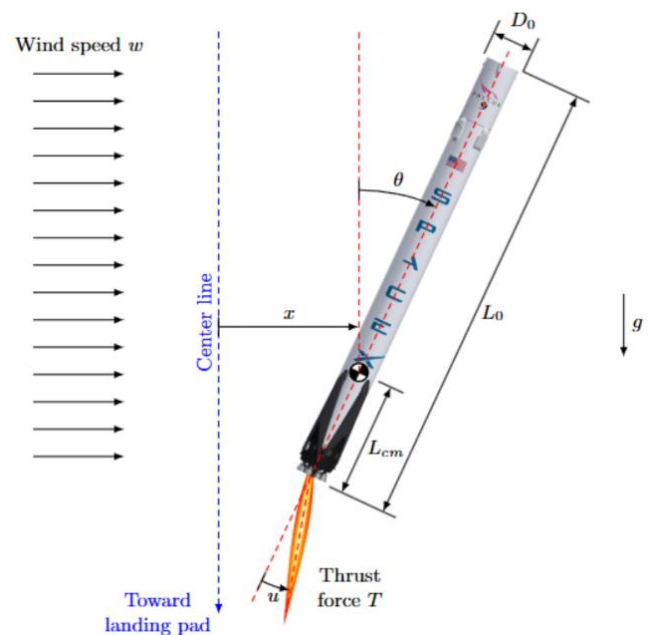
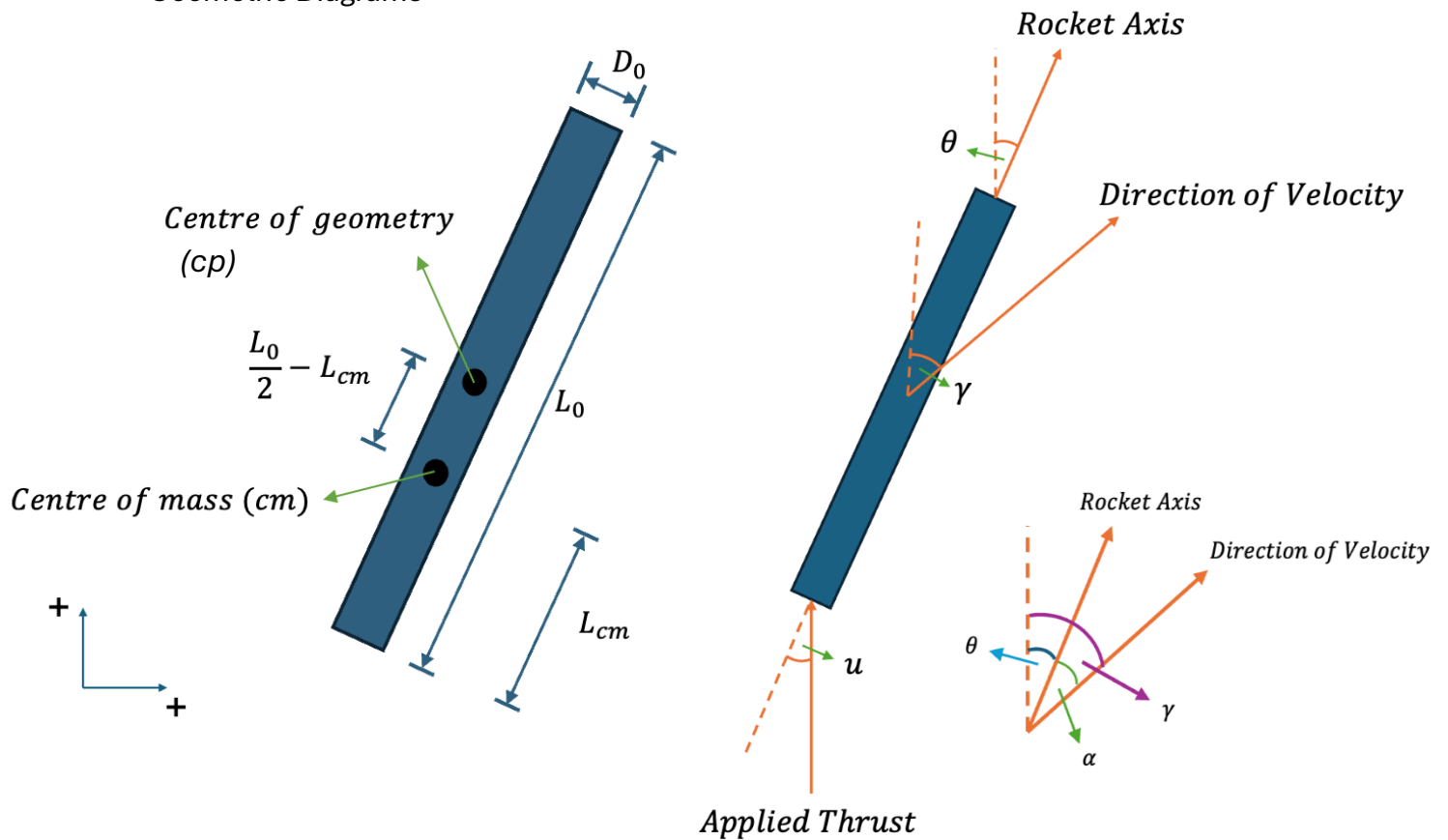


Figure 1: Relevant dimensions and signals for the Falcon 9's first stage.

Assumptions: One-dimensional motion (y), air resistance only present in the form of drag, no external disturbances, and constant parameters.

Modelling Equations of Motion (a):

Geometric Diagrams



Figures (2, 3): Geometric diagrams of the Falcon 9 rocket displaying relevant points, lengths, angles, and vectors.

From these diagrams the following variables can be derived:

- θ – The angle between the rocket axis and the vertical axis.
- α – The angle of attack or the angle between the rocket axis and the direction of velocity, defined as:

$$\alpha = \gamma - \theta$$

- v – Velocity vector, defined as:

$$v = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}$$

With its magnitude given by:

$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

- γ – The angle between the velocity vector and the vertical axis.
Also defined by:

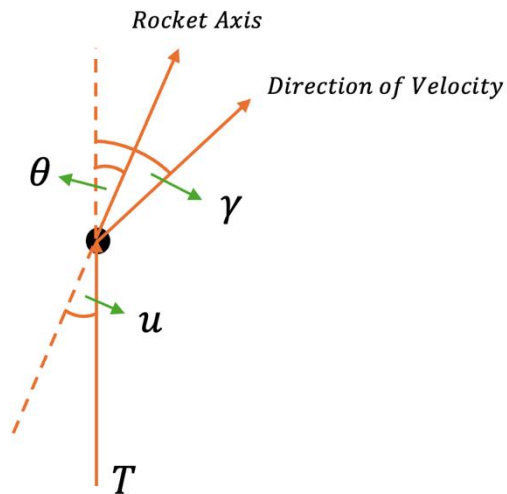
$$\gamma = \text{atan2}(\dot{x}, \dot{y})$$

- cm – Centre of mass
- cp – Centre of geometry or centre of pressure

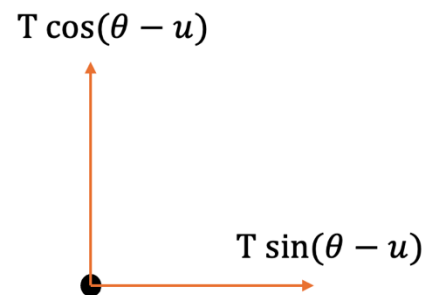
Free Body Diagrams:

Splitting up force components:

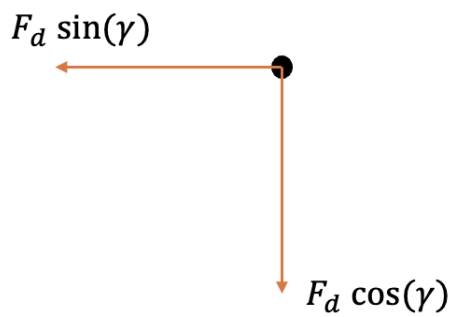
Thrust, T



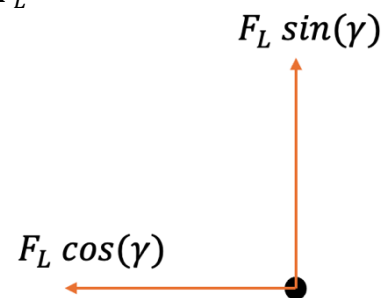
Splitting up into x and y,



Drag, F_d



Lift, F_L



Gravity



Figures (4-8): Angle and vector diagrams showing the directions in which forces acting on the Falcon 9 rocket will be applied.

Combining these:

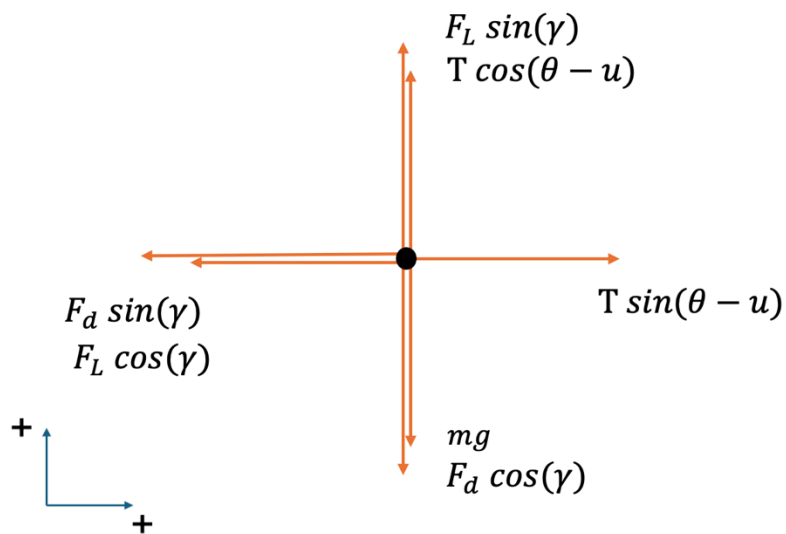


Figure 9: Diagram displaying the number of forces acting at a point on the Falcon 9

Applying $\sum F_x = m\ddot{x}$ and $\sum F_y = m\ddot{y}$

$$m\ddot{x} = T \sin(\theta - u) - F_L \cos(\gamma) - F_d \sin(\gamma)$$

$$m\ddot{y} = T \cos(\theta - u) + F_L \sin(\gamma) - F_d \cos(\gamma) - mg$$

Deriving a FBD for moments only including relevant forces for moments about cm:

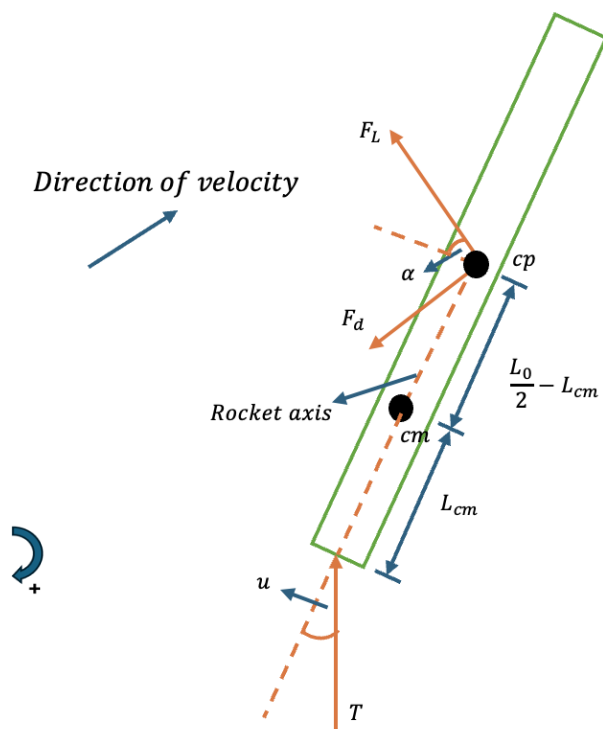


Figure 10: Diagram showing points at which forces act along the falcon 9, as well as relevant lengths, angles, and points.

Applying $\sum M = J\ddot{\theta}$

$$J\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right)(F_L \cos(\alpha) + F_d \sin(\alpha)) + TL_{cm} \sin(u)$$

Therefore, deriving the equations for the motion of the Rocket as:

$$m\ddot{x} = T \sin(\theta - u) - F_L \cos(\gamma) - F_d \sin(\gamma) \quad (\text{Eq. 1})$$

$$m\ddot{y} = T \cos(\theta - u) + F_L \sin(\gamma) - F_d \cos(\gamma) - mg \quad (\text{Eq. 2})$$

$$J\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right)(F_L \cos(\alpha) + F_d \sin(\alpha)) + TL_{cm} \sin(u) \quad (\text{Eq. 3})$$

Along with the given equations for drag and lift forces:

$$F_L = \frac{1}{2} \rho v^2 (A_n C_n - A_a C_a) \sin(\alpha) \cos(\alpha)$$

$$F_d = \frac{1}{2} \rho v^2 (A_a C_a \cos^2(\alpha) + A_n C_n \sin^2(\alpha))$$

Physical & Geometric Explanations:

The angle α is defined as the angle at which drag and lift are affecting the rocket from the angle the rocket is facing. This angle determines how lift and drag act on the rocket. Whereas the angle γ is the direction the whole rocket is going, not just the direction it is pointed.

The distance $\frac{L_0}{2} - L_{cm}$ appears because both lift and drag are acting on the rocket's midpoint but we take the moments about the centre of mass. This is just the distance between the centre of pressure or centre of the rocket and the centre of mass.

The term $\dot{x}^2 + \dot{y}^2$ appears because the rockets velocity depends on both the x and y velocity components, this is its velocity split into these components. $v^2 = \dot{x}^2 + \dot{y}^2$ being the full term.

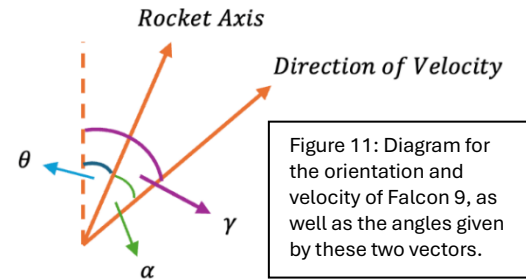
Equilibrium Points (b):

Expression for Thrust Force F_0 required to maintain constant vertical velocity:

$$m\ddot{x} = T\sin(\theta - u) - F_L\cos(\gamma) - F_d\sin(\gamma)$$

$$m\ddot{y} = T\cos(\theta - u) + F_L\sin(\gamma) - F_d\cos(\gamma) - mg$$

$$J\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right)(F_L\cos(\alpha) + F_d\sin(\alpha)) + TL_{cm}\sin(u)$$



When constant vertical speed,

Setting $T = T_0$, $\alpha = \gamma = 180^\circ$, $\theta = 0^\circ$, $u = 0$, $\dot{y} = \text{const}$, $\ddot{y} = 0$

Subbing in,

$$0 = T\sin(u) + F_L$$

$$0 = T_0 + F_d - mg$$

$$J\ddot{\theta} = \left(\frac{L_0}{2} - L_{cm}\right)(F_L) + TL_{cm}\sin(u)$$

Solving for T_0

$$T_0 = mg - F_d = mg - \frac{1}{2}\rho\dot{y}^2 A_a C_a \quad (\text{Eq. 4})$$

Reasoning:

When the equation is reduced, there is no lift or acceleration; Only drag, thrust and gravity are present.

Rocket's terminal velocity:

When the rocket's thrusters are off, $T_0 = 0$

$$0 = mg - F_d \Rightarrow F_d = mg$$

Given,

$$F_d = \frac{1}{2}\rho v^2 (A_a C_a \cos^2(\alpha) + A_n C_n \sin^2(\alpha))$$

$$mg = \frac{1}{2}\rho v^2 (A_a C_a \cos^2(\alpha) + A_n C_n \sin^2(\alpha))$$

Subbing in and rearranging $v = \dot{y}$, $\alpha = \gamma = 180^\circ$, $\theta = 0^\circ$, $u = 0$, $\dot{y} = \text{const}$, $\ddot{y} = 0$

$$mg = \frac{1}{2} \rho \dot{y}^2 (A_a C_a)$$

$$\dot{y} = \sqrt{\frac{mg}{\frac{1}{2} \rho A_a C_a}} \text{ (Eq. 5)}$$

Linearized EOM (c):

Linearizing the following equations of motion about the normal vertical descent equilibrium,

$$m\ddot{x} = T \sin(\theta - u) - F_L \cos(\gamma) - F_d \sin(\gamma)$$

$$J\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right) (F_L \cos(\alpha) + F_d \sin(\alpha)) + T L_{cm} \sin(u)$$

We are linearizing about steady vertical descent, so:

$$x, \dot{x}, \theta, \dot{\theta}, u = 0$$

$$\dot{y} = \dot{y}_0 = \text{const}$$

$$\ddot{y}_0 = 0$$

Linearization of angles:

First Linearizing α ,

$$\alpha_0 = \gamma_0 - \theta_0$$

$$\alpha_0 + \delta\alpha = \gamma_0 + \delta\gamma - (\theta_0 + \delta\theta)$$

But $\alpha_0 = \gamma_0 = \pi$, $\theta_0 = 0$,

Taking small perturbations gives,

$$\delta\alpha = \delta\gamma - \delta\theta$$

Then Linearizing γ ,

$$\gamma = \pi + \frac{\dot{x}}{\dot{y}_0}$$

$$\gamma_0 = \pi + \frac{\dot{x}_0}{\dot{y}_0}$$

$$\dot{x}_0 = 0,$$

$$\gamma_0 = \pi$$

$$\delta\gamma = \frac{\delta\dot{x}}{\dot{y}_0}$$

$$\alpha_0 = \pi$$

$$\delta\alpha = \frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta$$

Since the rocket is descending vertically, the velocity vector points downwards, verifying this.

Linearization of forces

Linearizing F_L :

$$F_{L0} = \frac{1}{2}\rho(\dot{x}_0^2 + \dot{y}_0^2)(A_n C_n - A_a C_a) \sin(\alpha_0) \cos(\alpha_0)$$

$$F_{L0} = \frac{1}{2}\rho(\dot{y}_0^2)(A_n C_n - A_a C_a) \sin(\alpha_0) \cos(\alpha_0)$$

Applying linearization:

$$F_{L0} + \delta F_L = \frac{1}{2}\rho(\dot{y}_0^2)(A_n C_n - A_a C_a) \sin(\alpha_0) \cos(\alpha_0) + \frac{\partial F_L}{\partial \alpha} \delta\alpha + \frac{\partial F_L}{\partial x} \delta x$$

$$\frac{\partial F_L}{\partial \alpha} = \frac{1}{2}\rho(\dot{y}_0^2)(A_n C_n - A_a C_a)(\cos^2(\alpha_0) - \sin^2(\alpha_0))$$

$$\cos^2(\alpha_0) - \sin^2(\alpha_0) = \cos(2\alpha_0),$$

At equilibrium:

$$\alpha_0 = \pi, \quad \cos(2\pi) = 1$$

$$\frac{\partial F_L}{\partial \alpha} = \frac{1}{2}\rho(\dot{y}_0^2)(A_n C_n - A_a C_a)(1)$$

$$\frac{\partial F_L}{\partial x} = \rho(\dot{x}_0)(A_n C_n - A_a C_a) \sin(\alpha_0) \cos(\alpha_0)$$

Since:

$$F_{L0} = \frac{1}{2}\rho(\dot{y}_0^2)(A_n C_n - A_a C_a) \sin(\alpha_0) \cos(\alpha_0),$$

they cancel, and since:

$$\dot{x}_0 = 0, \frac{\partial F_L}{\partial x} = 0,$$

It thus follows:

$$\delta F_L = \frac{\partial F_L}{\partial \alpha} \delta \alpha$$

$$\delta F_L = \frac{1}{2} \rho \dot{y}_0^2 (A_n C_n - A_a C_a) \delta \alpha$$

Linearizing F_d :

$$F_{d0} = \frac{1}{2} \rho [\dot{x}_0^2 A_a C_a \cos^2(\alpha_0) + \dot{x}_0^2 A_n C_n \sin^2(\alpha_0) + \dot{y}_0^2 A_a C_a \cos^2(\alpha_0) + \dot{y}_0^2 A_n C_n \sin^2(\alpha_0)]$$

$$\dot{x}_0 = 0 \text{ and } \alpha_0 = \pi,$$

$$F_{d0} = \frac{1}{2} \rho \dot{y}_0^2 [A_a C_a \cos^2(\alpha_0) + A_n C_n \sin^2(\alpha_0)] = \frac{1}{2} \rho \dot{y}_0^2 A_a C_a$$

$$F_{d0} + \delta F_d = \frac{1}{2} \rho \dot{y}_0^2 [A_a C_a \cos^2(\alpha_0) + A_n C_n \sin^2(\alpha_0)] + \frac{\partial F_d}{\partial \alpha} \delta \alpha + \frac{\partial F_d}{\partial x} \delta x$$

$\frac{\partial F_d}{\partial x} = 0$, since there is no x term in $F_{d0} = \frac{1}{2} \rho \dot{y}_0^2 [A_a C_a \cos^2(\alpha_0) + A_n C_n \sin^2(\alpha_0)]$, these terms cancel giving:

$$\delta F_d = \frac{\partial F_d}{\partial \alpha} \delta \alpha$$

With,

$$\frac{\partial F_d}{\partial \alpha} = \frac{1}{2} \rho \dot{y}_0^2 [A_a C_a \cos^2(\alpha_0) + A_n C_n \sin^2(\alpha_0)]$$

Therefore giving:

$$\delta F_d = \frac{1}{2} \rho \dot{y}_0^2 [A_a C_a (-2 \sin(\alpha_0) \cos(\alpha_0)) + A_n C_n (2 \sin(\alpha_0) \cos(\alpha_0))] \delta \alpha$$

$$\delta F_d = \rho \dot{y}_0^2 \sin(\alpha_0) \cos(\alpha_0) (A_n C_n - A_a C_a) \delta \alpha$$

Since $\sin(\alpha_0) = \sin(\pi) = 0$, $\delta F_d = 0$

Linearizing equations of motion:

Linearizing $\ddot{x}$:

$$m\ddot{x} = T\sin(\theta - u) - F_L\cos(\gamma) - F_d\sin(\gamma)$$

$$m\ddot{x}_0 + m\delta\ddot{x} = T_0\sin(\theta_0 - u_0) - F_{L0}\cos(\gamma_0) - F_{d0}\sin(\gamma_0) \\ + \frac{\partial f}{\partial F_L}\delta F_L + \frac{\partial f}{\partial F_d}\delta F_d + \frac{\partial f}{\partial \gamma}\delta\gamma + \frac{\partial f}{\partial \theta}\delta\theta + \frac{\partial f}{\partial u}\delta u$$

$$m\delta\ddot{x} = \frac{\partial f}{\partial F_L}\delta F_L + \frac{\partial f}{\partial F_d}\delta F_d + \frac{\partial f}{\partial \gamma}\delta\gamma + \frac{\partial f}{\partial \theta}\delta\theta + \frac{\partial f}{\partial u}\delta u$$

$$m\delta\ddot{x} = -\cos(\gamma_0)\delta F_L - \sin(\gamma_0)\delta F_d + (F_{L0}\sin(\gamma_0) - F_{d0}\cos(\gamma_0))\delta\gamma \\ + T_0\cos(\theta_0 - u_0)\delta\theta - T_0\cos(\theta_0 - u_0)\delta u$$

Plugging in $\gamma_0 = \pi$ gives:

$$m\delta\ddot{x} = \delta F_L + F_{d0}\delta\gamma + T_0\cos(\theta_0 - u_0)\delta\theta - T_0\cos(\theta_0 - u_0)\delta u$$

Plugging in $\theta = 0$, we get:

$$m\delta\ddot{x} = \delta F_L + F_{d0}\frac{\delta x}{\dot{y}_0} + T_0\delta\theta - T_0\delta u$$

Subbing in:

$$\delta F_L = \frac{1}{2}\rho\dot{y}_0^2(A_nC_n - A_aC_a)\delta\alpha = \frac{1}{2}\rho\dot{y}_0^2(A_nC - A_aC_a)\left(\frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta\right)$$

Gives us:

$$m\delta\ddot{x} = \frac{1}{2}\rho\dot{y}_0^2(A_nC_n - A_aC_a)\left(\frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta\right) + F_{d0}\frac{\delta\dot{x}}{\dot{y}_0} + T_0\delta\theta - T_0\delta u$$

And subbing in:

$$F_{d0} = \frac{1}{2}\rho\dot{y}_0^2 A_aC_a$$

Gives:

$$m\delta\ddot{x} = \frac{1}{2}\rho\dot{y}_0^2(A_nC_n - A_aC_a)\left(\frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta\right) + \frac{1}{2}\rho\dot{y}_0^2 A_aC_a\frac{\delta\dot{x}}{\dot{y}_0} + T_0\delta\theta - T_0\delta u$$

Rearranged collecting terms:

$$m\delta\ddot{x} = \frac{1}{2}\rho\dot{y}_0(A_nC_n - A_aC_a)\delta\dot{x} - \frac{1}{2}\rho\dot{y}_0^2(A_nC_n - A_aC_a)\delta\theta + \frac{1}{2}\rho\dot{y}_0 A_aC_a\delta\dot{x} + T_0\delta\theta - T_0\delta u$$

Combining like terms, gives fully linearized form:

$$m\delta\ddot{x} = \frac{1}{2}\rho\dot{y}_0 A_n C_n \delta\dot{x} + \left(T_0 - \frac{1}{2}\rho\dot{y}_0^2 (A_n C_n - A_a C_a) \right) \delta\theta - T_0 \delta u$$

Linearizing $\ddot{\theta}$:

$$J\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right) (F_L \cos(\alpha) + F_d \sin(\alpha)) + T L_{cm} \sin(u)$$

$$J(\delta\ddot{\theta}) = \frac{\partial f}{\partial F_L} \delta F_L + \frac{\partial f}{\partial F_D} \delta F_D + \frac{\partial f}{\partial \alpha} \delta \alpha + \frac{\partial f}{\partial u} \delta u$$

$$J\delta\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right) [\delta F_L \cos \alpha_0 + \delta F_d \sin \alpha_0 + (-F_{L0} \sin \alpha_0 + F_{d0} \cos \alpha_0) \delta \alpha] \\ + T_0 L_{cm} \cos u_0 \delta u$$

Plugging in $\alpha_0 = \pi$, $\delta F_d = 0$, and $u_0 = 0$ gives:

$$J\delta\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right) [\delta F_L + (F_{d0}) \delta \alpha] + T_0 L_{cm} \delta u$$

$$\delta \alpha = \frac{\delta \dot{x}}{\dot{y}_0} - \delta \theta \\ F_{d0} = \frac{1}{2} \rho \dot{y}_0^2 A_a C_a \\ \delta F_L = \frac{1}{2} \rho \dot{y}_0^2 (A_n C_n - A_a C_a) \left(\frac{\delta \dot{x}}{\dot{y}_0} - \delta \theta \right)$$

Subbing these in:

$$J\delta\ddot{\theta} = -\left(\frac{L_0}{2} - L_{cm}\right) \left[\frac{1}{2} \rho \dot{y}_0^2 (A_n C_n - A_a C_a) \left(\frac{\delta \dot{x}}{\dot{y}_0} - \delta \theta \right) + \frac{1}{2} \rho \dot{y}_0^2 A_a C_a \left(\frac{\delta \dot{x}}{\dot{y}_0} - \delta \theta \right) \right] \\ + T_0 L_{cm} \delta u$$

Rearranging and combining like terms gives fully linearized form:

$$J\delta\ddot{\theta} = -\frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2} \frac{1}{2} \rho \dot{y}_0 A_n C_n \delta\dot{x} + \frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2} \rho \dot{y}_0^2 A_n C_n \delta\theta + T_0 L_{cm} \delta u$$

Otherwise expressed as:

$$J\delta\ddot{\theta} = -\frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2} \rho \dot{y}_0^2 (A_n C_n) \left(\frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta\right) + T_0 L_{cm} \delta u$$

The final forms of the linearized equations are as follows:

$$m\delta\ddot{x} = \frac{1}{2} \rho \dot{y}_0 A_n C_n \delta\dot{x} + \left(T_0 - \frac{1}{2} \rho \dot{y}_0^2 (A_n C_n - A_a C_a)\right) \delta\theta - T_0 \delta u \quad (Eq. 6)$$

$$J\delta\ddot{\theta} = -\frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2} \rho \dot{y}_0^2 (A_n C_n) \left(\frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta\right) + T_0 L_{cm} \delta u \quad (Eq. 7)$$

Simulink Implementation (d):

Using provided rocket_landing.slx file

- (i) Implement ODEs from (a) as a subsystem in Simulink
- (ii) Implement linearized ODEs from (c) as a subsystem in Simulink (should be the same as (i))
- (iii) Test and tabulate the simulation with ICs given for both nonlinearized and linearized systems. How different are they and what if much larger theta(0) is used.

In the initial instructions for part (a), several non-linear ordinary differential equations were provided for the forces, directions, and angles influencing the rocket's translation. These equations were inserted as a subsystem in Simulink, with input variables for thrust, thrust angle, and velocity in the y direction. $(T, u, \dot{y})$, and output variables for vertical position and vertical alignment (x, θ) , respectively. These non-linear ODEs provided from part (a) are modelled by:

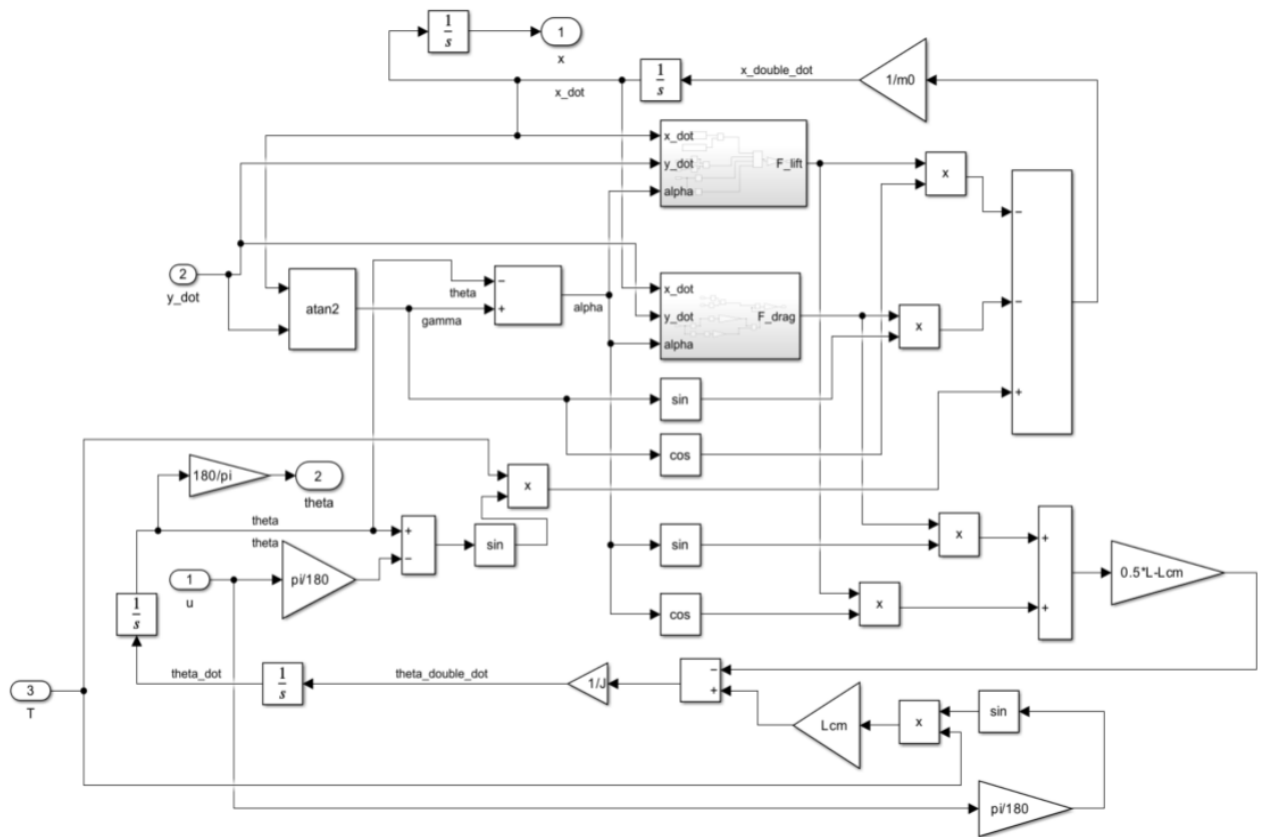


Figure 12: Nonlinear model Simulink block diagram

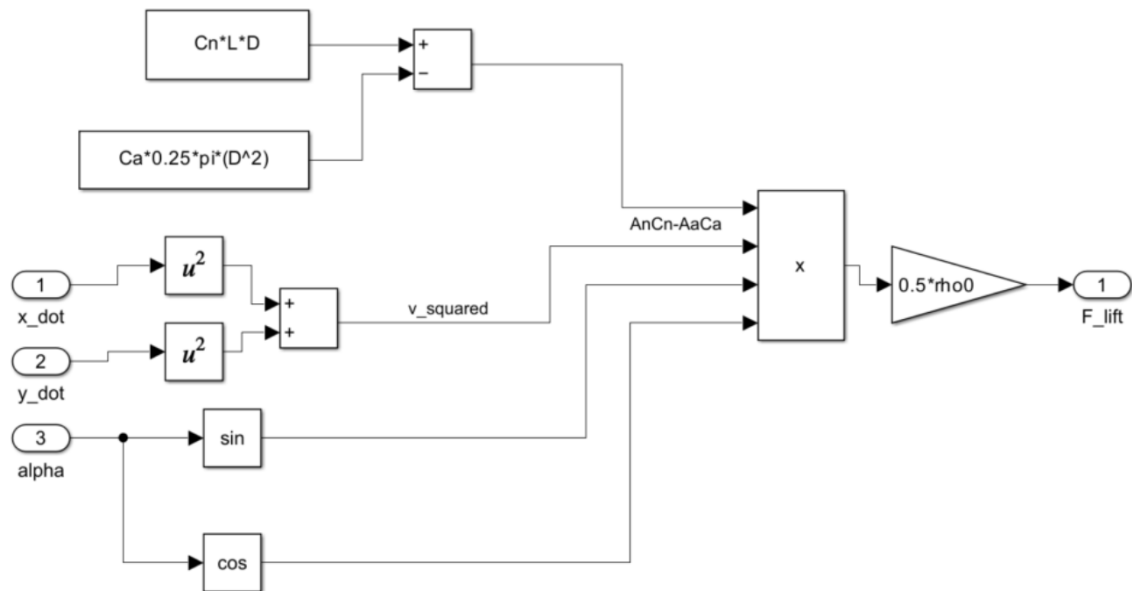


Figure 13: Nonlinear model lift force subsystem

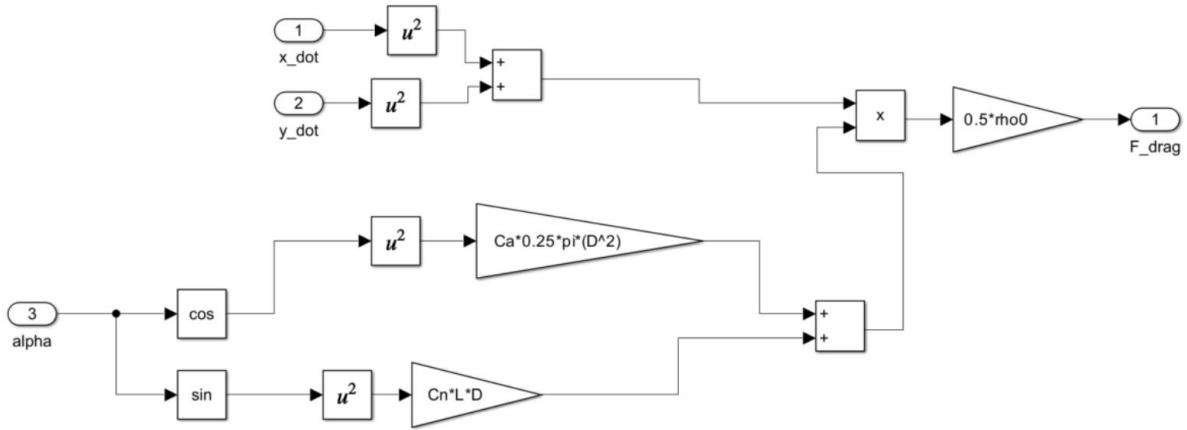


Figure 14: Nonlinear model drag force subsystem

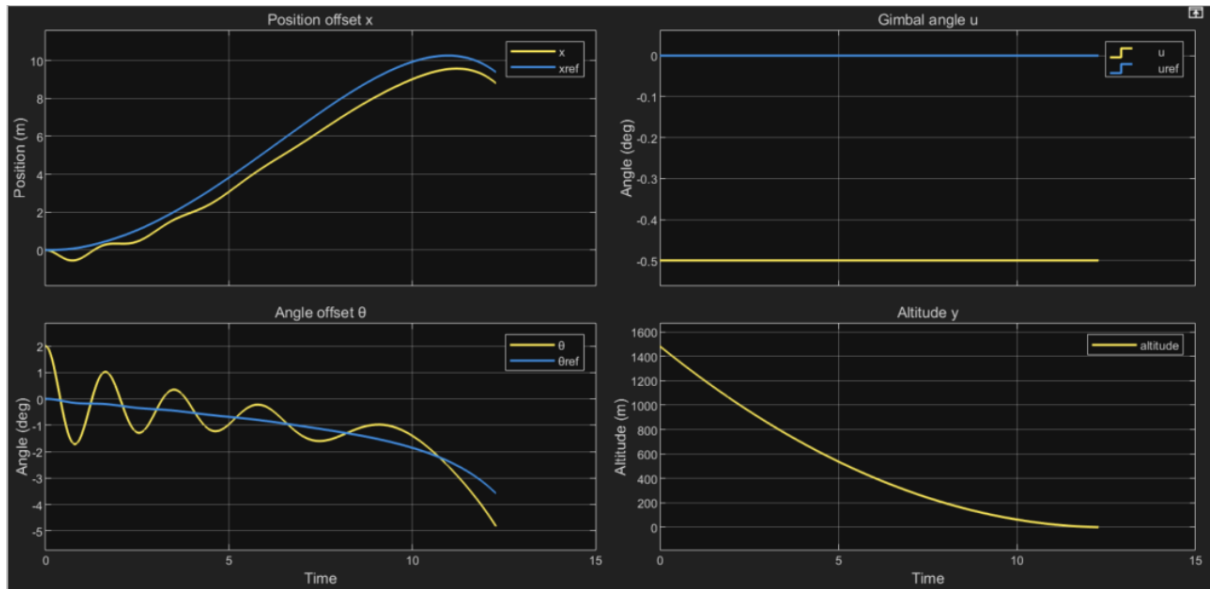


Figure 15: Nonlinear model (blue) vs true model (yellow) graphs, with initial conditions $\theta(0) = 2 \text{ degrees}$, $\dot{y}(0) = -236 \text{ m/s}$

The nonlinear model and the true model share some characteristics. The first being a close resemblance in behaviours for their respective “Position offset x ” graphs. Since the true model accounts for change in air density, mass, and polar moment of inertia, this results in a slight wave behaviour at the beginning of the graph. As for the “Angle offset θ ” graphs, the general behaviour is captured in the model, however the oscillations are not.

Following the modelling of the nonlinear motion equations in part (a), linearized forms of the equations were then modelled with the same previous inputs and outputs and then compared against their counterparts. The resulting block diagram and graph are modelled below, demonstrating a test with the inputs of: $(x(0) = 0, \theta(0) = 2^\circ, \dot{x}(0) = 0, \dot{\theta}(0) = 0, \dot{y}(0) = -236 \frac{m}{s}, u(0) = -0.5^\circ)$:

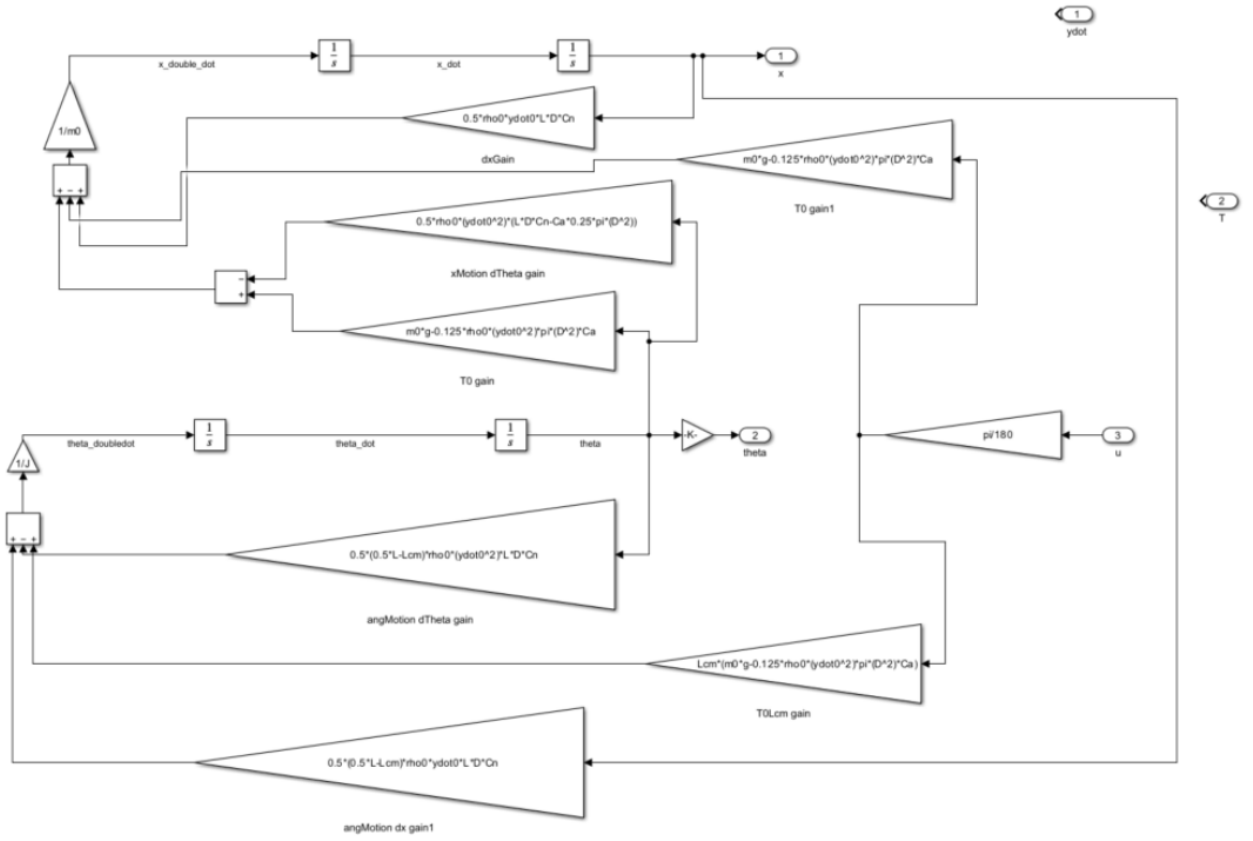


Figure 16: Linearized model Simulink block diagram

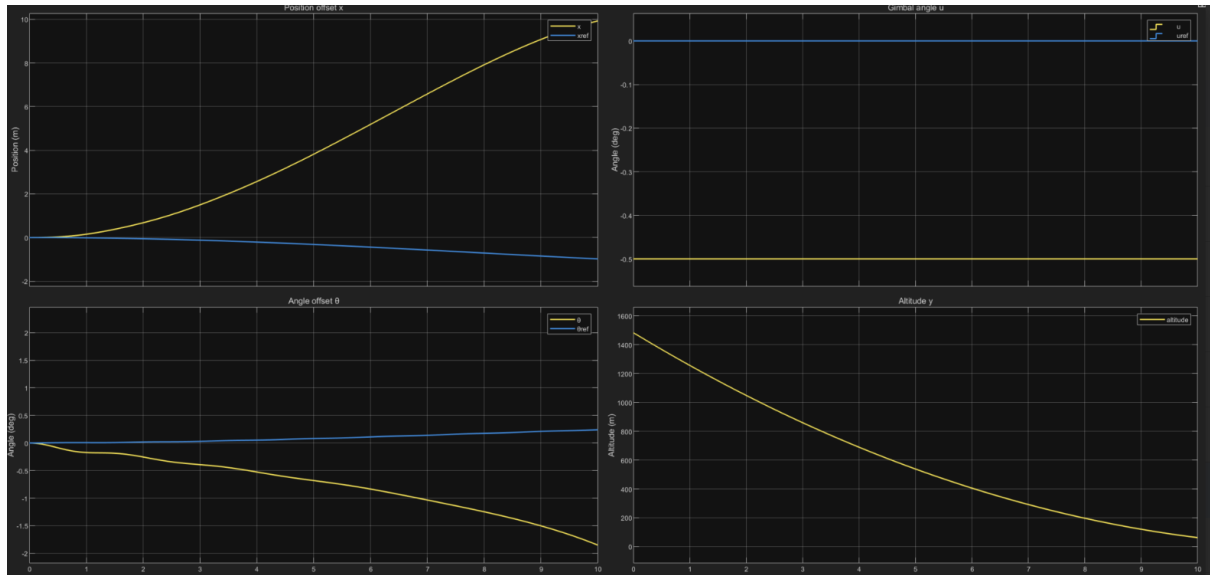


Figure 17: Nonlinear model (yellow) vs linear model (blue) graphs, with initial conditions $\theta(0) = 2 \text{ degrees}$, $\dot{y}(0) = -236 \text{ m/s}$

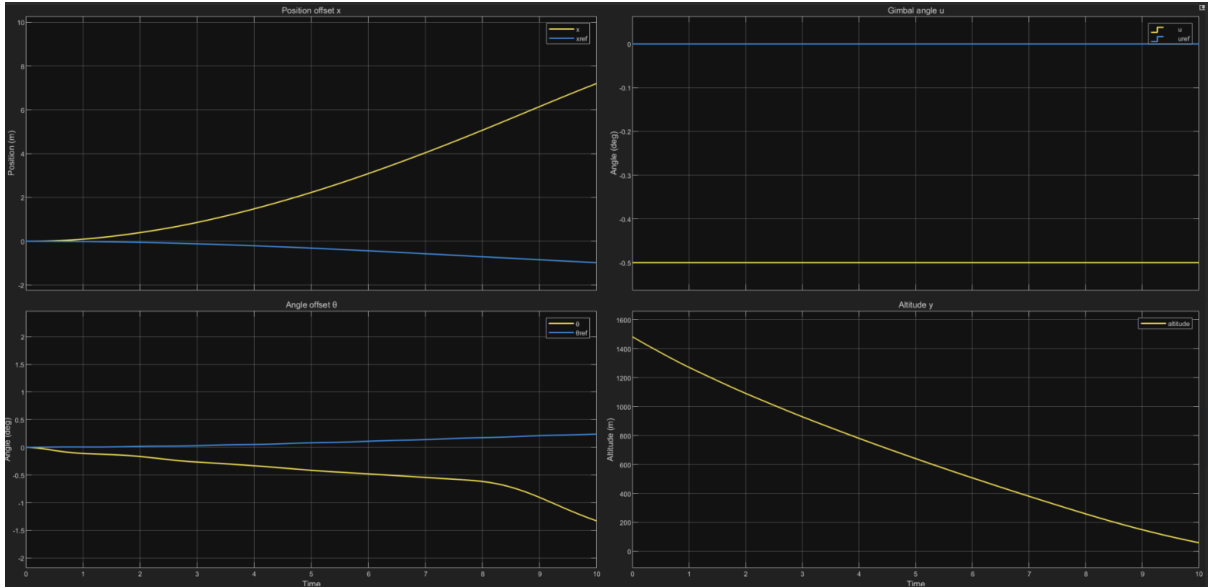


Figure 18: Nonlinear model (yellow) vs linear model (blue) graphs, with initial conditions $\theta(0) = 45 \text{ degrees}$, $\dot{y}(0) = -236 \text{ m/s}$

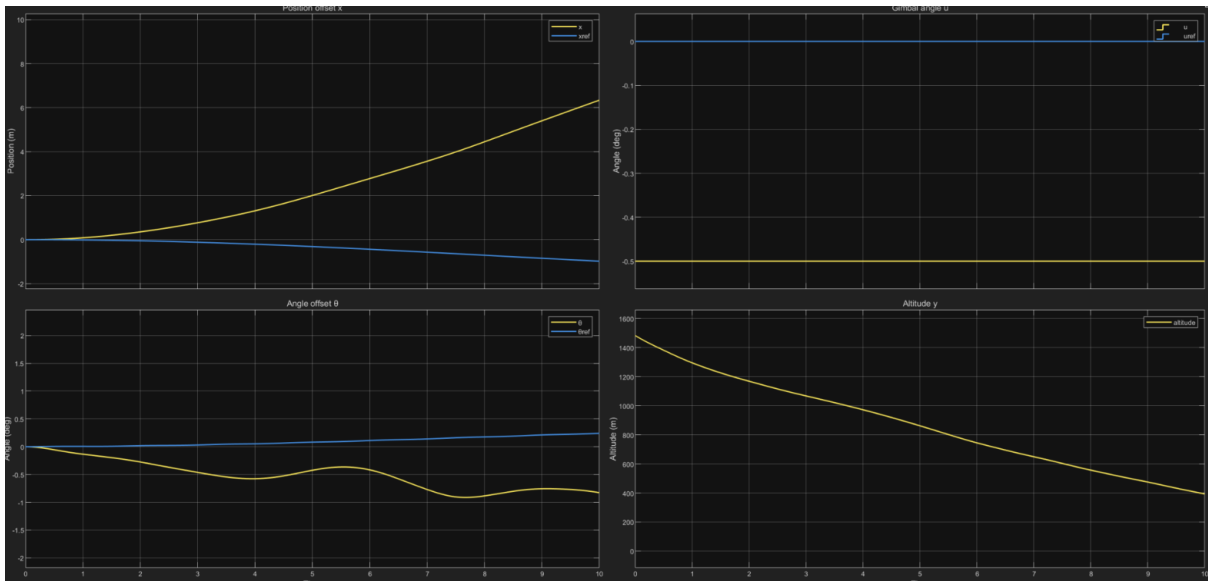


Figure 19: Nonlinear model (yellow) vs linear model (blue) graphs, with initial conditions $\theta(0) = 90 \text{ degrees}$, $\dot{y}(0) = -236 \text{ m/s}$

Graph analysis:

From the above graphs, it can be observed that there is a lack of correlation between the nonlinear model and the true model regardless of the initial angle given. This likely due to potential errors with the linearized model, as well as further errors presented by an inaccurately determined equilibrium point.

Stability Analysis (e)

Linearized model simulation

- (i) Simulate non-linear system for conditions given and comment on stability

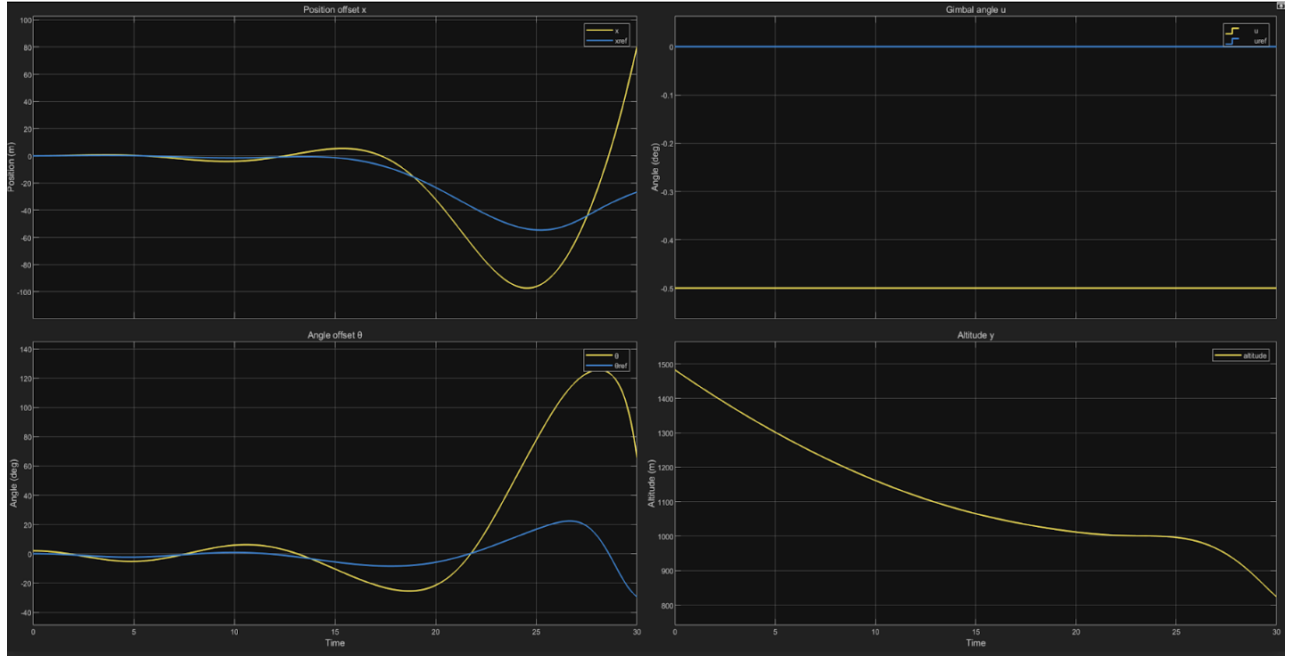


Figure 20: Nonlinear model vs true model graphs, with initial conditions $\theta(0) = 2 \text{ degrees}$, $\dot{y}(0) = -40 \text{ m/s}$

Based on the graphs shown in Figure 20, the system indicates an unstable motion, due to the increasing amplitudes of the x and theta values with respect to time.

- (ii) Compute transfer function for with linearized model from (c), then apply stability criteria to find critical descent speed where rocket goes from being stable to unstable (symbolic answer). Based on results explain why large values of T_0 require the rocket to fall faster to ensure stability.

Beginning with the equations found in part c:

$$m\delta\ddot{x} = \frac{1}{2}\rho y_0 A_n C_n \delta\dot{x} + \left(T_0 - \frac{1}{2}\rho \dot{y}_0^2 (A_n C_n - A_a C_a)\right) \delta\theta - T_0 \delta u \quad (\text{Eq. 6})$$

$$J\delta\ddot{\theta} = -\frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2} \rho \dot{y}_0^2 (A_n C_n) \left(\frac{\delta\dot{x}}{\dot{y}_0} - \delta\theta\right) + T_0 L_{cm} \delta u \quad (\text{Eq. 7})$$

the goal is to find the transfer function between θ and u . Starting with motion in the x-direction, the differential equation is converted into the Laplace domain, and like terms are combined. $X(s)$ is then isolated to be utilized in the angular motion equation.

$$\left(ms^2 - \frac{1}{2}\rho y_0 A_n C_n s\right) X(s) = \left(T_0 - \frac{1}{2}\rho \dot{y}_0^2 (A_n C_n - A_a C_a)\right) \theta(s) - T_0 U(s)$$

$$X(s) = \frac{\left(T_0 - \frac{1}{2}\rho\dot{y}_0^2(A_n C_n - A_a C_a)\right)\theta(s) - T_0 U(s)}{\left(ms^2 - \frac{1}{2}\rho\dot{y}_0 A_n C_n s\right)}$$

Converting the angular motion equation to the Laplace domain and plugging in X(s) gives the following.

$$Js^2\theta(s) = -\frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2}\rho\dot{y}_0(A_n C_n) \left(\frac{\left(T_0 - \frac{1}{2}\rho\dot{y}_0^2(A_n C_n - A_a C_a)\right)\theta(s) - T_0 U(s)}{\left(ms^2 - \frac{1}{2}\rho\dot{y}_0 A_n C_n s\right)} \right) s$$

$$- \frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2}\rho\dot{y}_0^2(A_n C_n)\theta(s) + T_0 L_{cm} U(s)$$

Combining like terms, with $U(s)$ on the left side of the equation and $\theta(s)$ on the right side of the equation gives the following equation below. Additionally, since the angular motion equation contained a first order derivative term of x, this results in a cancelation of s terms in the numerator and denominator.

$$\frac{T_0 \left(\frac{L_0}{2} - L_{cm}\right) \rho\dot{y}_0(A_n C_n) - 2T_0 L_{cm} \left(ms - \frac{1}{2}\rho\dot{y}_0 A_n C_n\right)}{2 \left(ms - \frac{1}{2}\rho\dot{y}_0 A_n C_n\right)} U(s)$$

$$= \left[-Js^2 - \frac{\left(\frac{L_0}{2} - L_{cm}\right)}{2}\rho\dot{y}_0^2(A_n C_n) \right. \\ \left. + \frac{\left(\frac{L_0}{2} - L_{cm}\right) \rho\dot{y}_0(A_n C_n) \left(T_0 - \frac{1}{2}\rho\dot{y}_0^2(A_n C_n - A_a C_a)\right)}{2 \left(ms - \frac{1}{2}\rho\dot{y}_0 A_n C_n\right)} \right] \theta(s)$$

Setting the common denominator equal to $2 \left(ms - \frac{1}{2}\rho\dot{y}_0 A_n C_n\right)$ and rearranging terms gives the following transfer function. This leads the numerator terms on the right side of the equation to now be on the denominator of the transfer function. The numerator terms on the left side of the equation stay as the numerator terms of the transfer function.

$$\frac{\theta(s)}{U(s)} = \frac{T_0 \left(\frac{L_0}{2} - L_{cm}\right) \rho\dot{y}_0(A_n C_n) - 2T_0 L_{cm} \left(ms - \frac{1}{2}\rho\dot{y}_0 A_n C_n\right)}{\left(ms - \frac{1}{2}\rho\dot{y}_0 A_n C_n\right) \left[-2Js^2 - \left(\frac{L_0}{2} - L_{cm}\right) \rho\dot{y}_0^2(A_n C_n) \right] + \left(\frac{L_0}{2} - L_{cm}\right) \rho\dot{y}_0(A_n C_n) \left(T_0 - \frac{1}{2}\rho\dot{y}_0^2(A_n C_n - A_a C_a)\right)}$$

Distributing terms and rearranging both the numerator and denominator into its characteristic polynomial form gives the following.

$$\frac{\theta(s)}{U(s)} = \frac{-2T_0L_{cm}ms + T_0\rho\dot{y}_0A_nC_n\left(\frac{L_0 - L_{cm}}{2}\right)}{-2Jms^3 + J\rho\dot{y}_0A_nC_ns^2 - m\rho\dot{y}_0^2A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)s + \rho\dot{y}_0A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)\left(\frac{1}{2}\rho\dot{y}_0^2A_nC_n + T_0\right)}$$

Normalizing the characteristic polynomial gives the final transfer function.

$$\frac{\theta(s)}{U(s)} = \frac{\frac{1}{J}T_0L_{cm}s - \frac{1}{2Jm}T_0\rho\dot{y}_0A_nC_n\left(\frac{L_0 - L_{cm}}{2}\right)}{s^3 - \frac{1}{2m}\rho\dot{y}_0A_nC_ns^2 + \frac{1}{2J}m\rho\dot{y}_0^2A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)s - \frac{1}{2Jm}\rho\dot{y}_0A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)\left(\frac{1}{2}\rho\dot{y}_0^2A_nC_n + T_0\right)}$$

(Eq. 8)

With the transfer function found in the standard form, the critical stability point of the system can be found by taking the characteristic polynomial and assessing the stability criteria based on its order.

$$s^3 - \frac{1}{2m}\rho\dot{y}_0A_nC_ns^2 + \frac{1}{2J}m\rho\dot{y}_0^2A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)s - \frac{1}{2Jm}\rho\dot{y}_0A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)\left(\frac{1}{2}\rho\dot{y}_0^2A_nC_n + T_0\right)$$

The critical stability point indicates for what value of $\dot{y}_0$ would the function shift from stable to unstable and vice versa. At first glance, two coefficients appear to be negative, rendering it impossible to determine the critical stability point. However, since $\dot{y}_0$ is negative, this converts the negative terms into positives (with the expression $\left(\frac{L_0}{2} - L_{cm}\right)$ also being a positive value. Thus, the stability criterion for a third order system is as follows:

$$a_2a_1 > a_0$$

where $a_2 = \frac{1}{2m}\rho\dot{y}_0A_nC_n$, $a_1 = \frac{1}{2J}m\rho\dot{y}_0^2A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)$, and $a_0 = \frac{1}{2Jm}\rho\dot{y}_0A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)\left(\frac{1}{2}\rho\dot{y}_0^2A_nC_n + T_0\right)$. This results in the following inequality.

$$\left[\frac{1}{2m}\rho\dot{y}_0A_nC_n\right] * \left[\frac{1}{2J}m\rho\dot{y}_0^2A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)\right] > \frac{1}{2Jm}\rho\dot{y}_0A_nC_n\left(\frac{L_0}{2} - L_{cm}\right)\left(\frac{1}{2}\rho\dot{y}_0^2A_nC_n + T_0\right)$$

Since the single critical stability point is needed, rather than a set of values to determine stability, the inequality can be set up as an equation to then solve for $\dot{y}_0$. Reducing the equation and combining in terms of orders of $\dot{y}_0$ results in the following.

$$\begin{aligned} \dot{y}_0^3 \left[\frac{1}{4J} (\rho A_n C_n)^2 \left(\frac{L_0}{2} - L_{cm} \right) \right] \\ = \dot{y}_0^3 \left[\frac{1}{4Jm} (\rho A_n C_n)^2 \left(\frac{L_0}{2} - L_{cm} \right) \right] + \frac{T_0}{2Jm} \rho A_n C_n \left(\frac{L_0}{2} - L_{cm} \right) \dot{y}_0 \end{aligned}$$

Dividing $\dot{y}_0$ on both sides of the equation and combining the remaining $\dot{y}_0$ terms on the left side results in the following.

$$\dot{y}_0^2 \left[\frac{1}{4J} (\rho A_n C_n)^2 \left(\frac{L_0}{2} - L_{cm} \right) \left(1 - \frac{1}{m} \right) \right] = \frac{T_0}{2Jm} \rho A_n C_n \left(\frac{L_0}{2} - L_{cm} \right)$$

Finally, dividing both sides by the $\dot{y}_0^2$ coefficient and taking the square of both sides results in the final critical value of $\dot{y}_0$.

$$\dot{y}_{crit} = \sqrt{\frac{2T_0}{\rho A_n C_n (m - 1)}} \quad (Eq. 9)$$

Since $\dot{y}_{crit}$ is dependent on T_0 , an increase in T_0 , results in an increase in $\dot{y}_{crit}$. This results in an increase in descent speed values greater than $\dot{y}_{crit}$ needed to maintain the stability of the system.

It can also be reasoned that the values must be greater than, not less than, the critical descent speed by analysing at $T_0 = 0$. Since this would result in a critical descent speed of 0 m/s, this would result in all values less than the critical speed to be negative. In this instance, since the rocket is descending, a negative value indicates that the rocket would be ascending. However, since there is no thrust, there is no force to result in an upward motion in the first place. Therefore, a stable system results when the values are greater than the $\dot{y}_{crit}$ value.